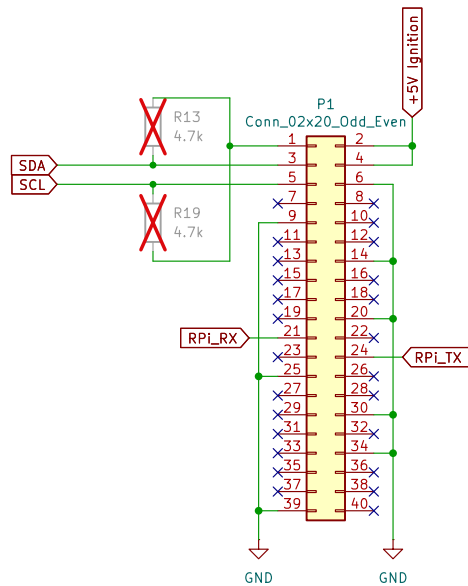




ESP32-WROOM

File: ESP32\_WROOM.kicad\_sch

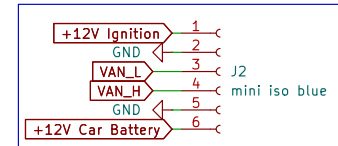
ECU\_comms

File: ECU\_comms.kicad\_sch

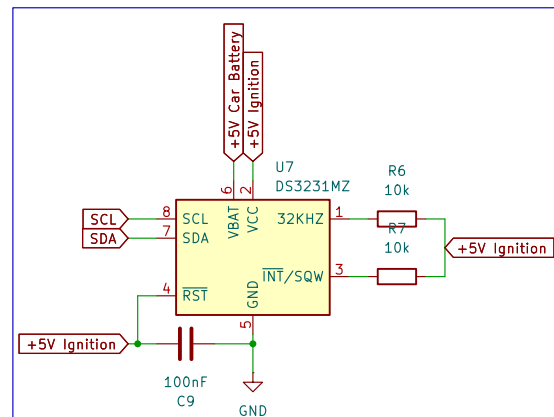
Power

File: power.kicad\_sch

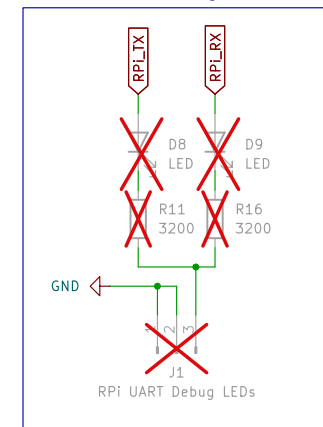
Mini ISO Blue -> JST connector



RPi RTC



UART Debug LEDs



This board is designed to act as a HAT for a Raspberry Pi 3B (non plus).

It would work for a 3B+, but the RUN header is not in the same place, so the power control would not work.

There are 2 power rails, one for the ESP, and one for the Pi. The ESP is designed to be powered by the permanent 12 volt rail from the car, and the Pi from the VAN+ rail that comes on when the car wakes up from sleep.

By using the `{f7473ed4-a3a0-4b2d-bc64-33b6e5e2e529:VALUE}` jumper, you can power the ESP from the VAN+ Power rail.

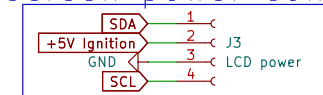
Note: The VAN+ power rail is also 12 volts, but this board expects it to be regulated by an external voltage regulator (for now).

The board can also be used as a standalone VAN bus reader.

Mounting Holes



LCD screen power connector



Sheet: /

File: main.kicad\_sch

**Title: Peugeot Project PCB**

Size: A4

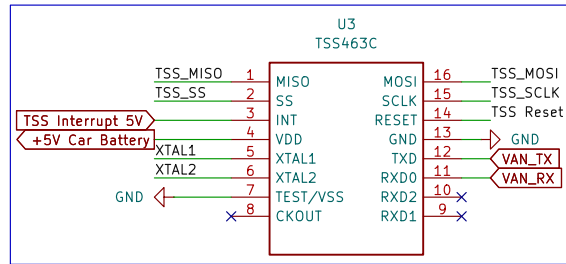
Date:

KiCad E.D.A. 9.0.4

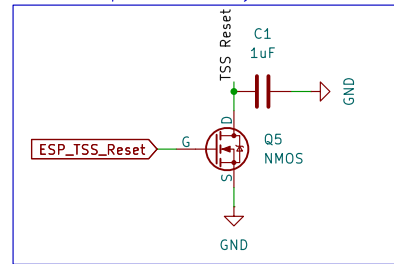
**Rev: v1.0**

Id: 1/4

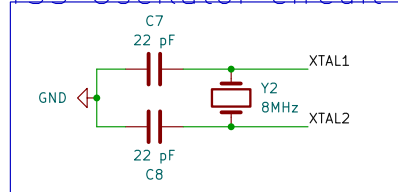
## TSS – VAN DLC



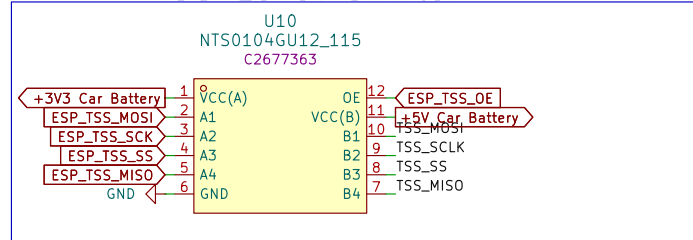
## TSS Reset pin driven by a 3V3 ESP pin



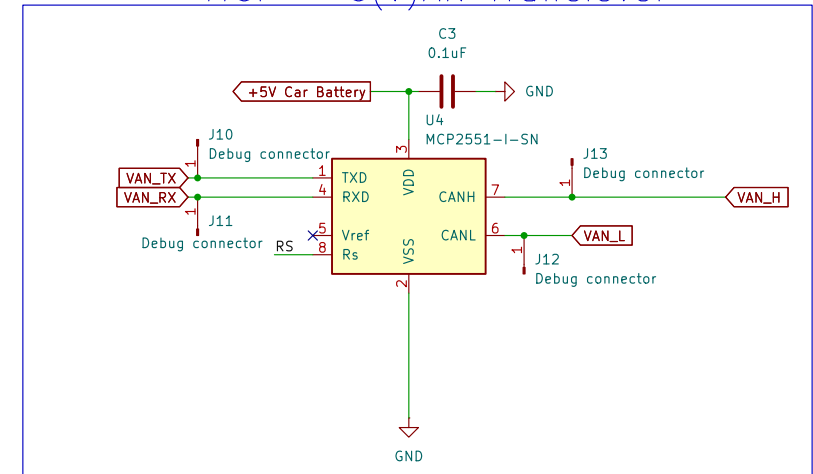
## TSS Oscillator circuit



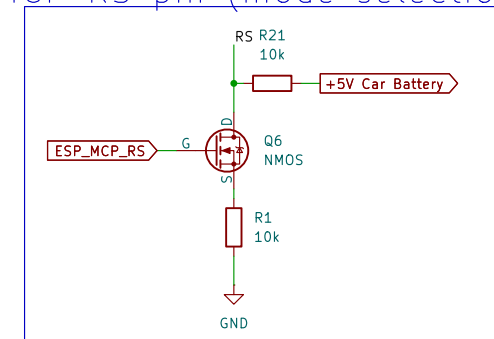
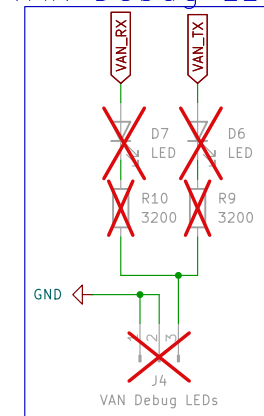
## TSS Level Shifter



## MCP – C(V)AN Tranciever



## VAN Debug LEDs MCP RS pin (mode selection)



Sheet: /ECU\_comms/  
File: ECU\_comms.kicad\_sch

**Title: Peugeot Project PCB**

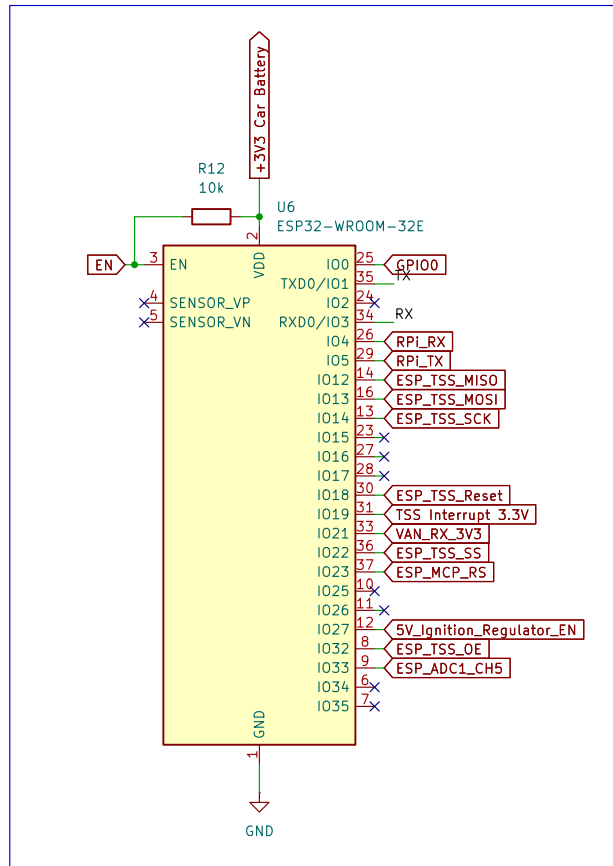
Size: A4  
KiCad E.D.A. 9.0.4

Date:

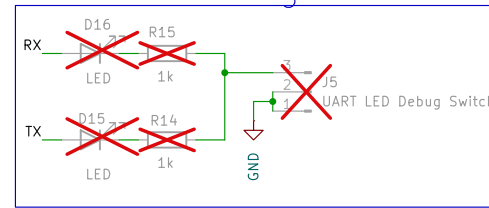
Rev: v1.0

Id: 2/4

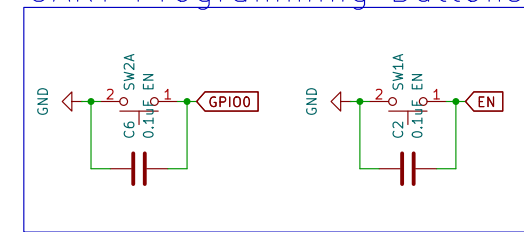
## ESP 32 WROOM 32E



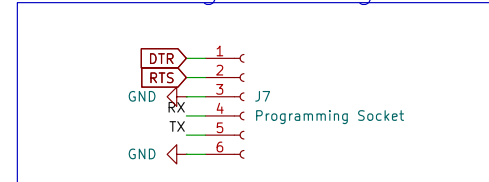
## UART Debug LEDs



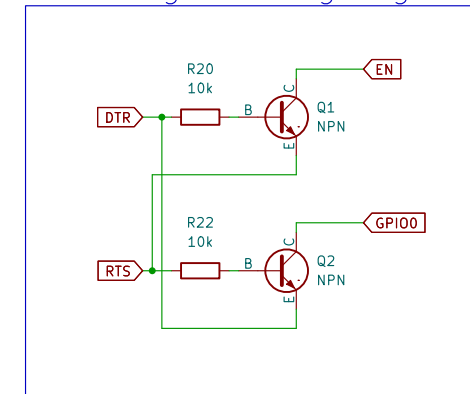
## UART Programming Buttons



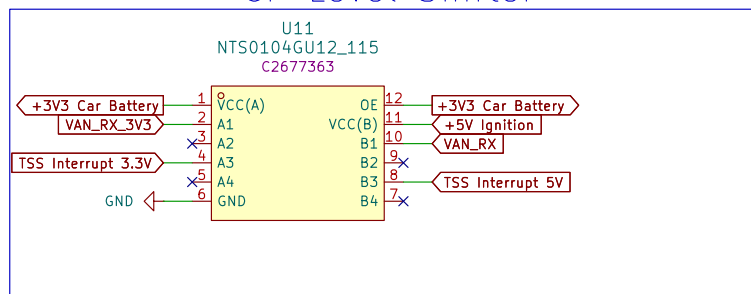
## ESP32 Programming Socket



## UART Programming Signals



## GP Level Shifter



Sheet: /ESP32-WROOM/  
File: ESP32\_WROOM.kicad\_sch

**Title: Peugeot Project PCB**

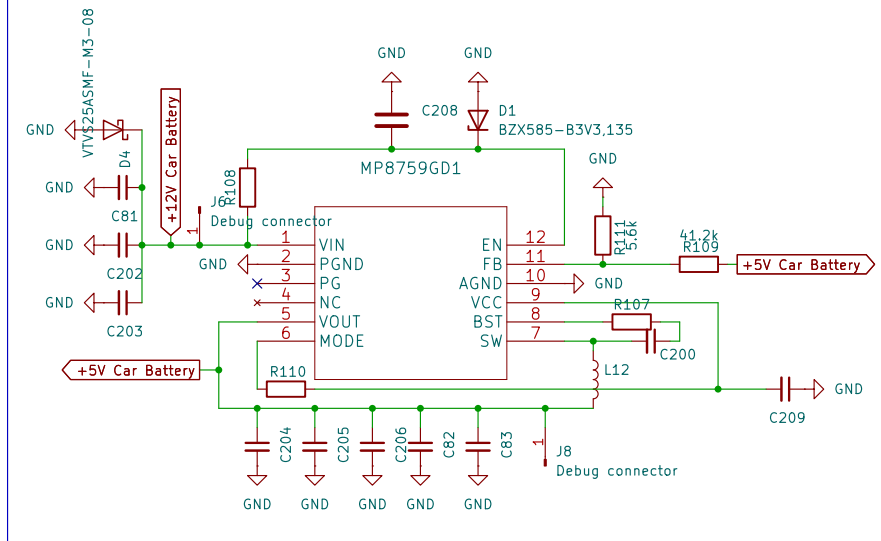
Size: A4 Date:

KiCad E.D.A. 9.0.4

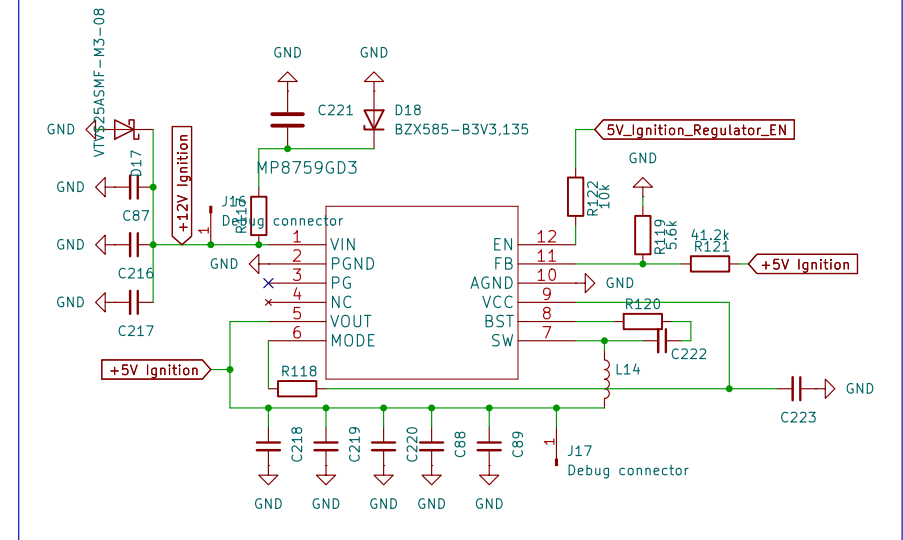
**Rev: v1.0**

Id: 3/4

## 5V Volts Power Regulator



## 5V Ignition Lock Power Regulator



## 3V3 Volts Power Regulator

